

Jamie Mahowald

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Education

Stanford University

Sep. 2027 – present

- PhD, Computational and Mathematical Engineering (CME)

The University of Texas at Austin

Aug. 2021 – May 2025

- BS, Mathematics, GPA: 3.98/4.00
- BA, Plan II Honors, minors in Latin and government

Positions

- **Post-bachelor student** | Los Alamos National Laboratory (LANL) May 2025 – July 2026
- **Student technician** | Applied Research Laboratories at UT Austin (ARL:UT) Aug. 2024 – May 2025
- **Research intern** | Oden Institute for Computational Engineering & Sciences Jun. – Aug. 2024
- **Research intern** | Library of Congress, John W. Kluge Center Jan. – Apr. 2024
- **Research intern** | Comparative Constitutions Project (www.constituteproject.org) May – Aug. 2023

Honors and awards

- Phi Beta Kappa, UT Austin. May 2025
- Special honors in Mathematics (senior honors thesis distinction), UT Austin. May 2025
- Kuhn Scholarship, UT Austin. Sep. 2024
- Best Long Paper, *Computational Humanities Research* (CHR). Oct. 2024
- Plan II Travel Grant, UT Austin. Jul. 2024
- Chris R. Bjornson Scholarship, Archer Center. Jan. 2024
- Plan II Continuing Scholarship, UT Austin. Oct. 2022

Publications

Peer-reviewed:

- J. Mahowald, B.C.G. Lee. "Retrieval-Augmented Search for Large-Scale Map Collections with ColPali." *ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR)*, 2025. <https://arxiv.org/abs/2510.25718>.
- J. Mahowald and T. Bui-Thanh. "Generalization Limits of In-Context Operator Networks for Higher-Order Partial Differential Equations." *SIAM Undergraduate Research Online*, 2025. <https://www.siam.org/media/t1xb33qb/s172206rr.pdf>.
- J. Mahowald, R. Srinivasan, and Z. Wang. "Toward Dataset Distillation for Regression Problems." *Efficient Systems for Foundation Models Workshop, International Conference on Machine Learning (ICML)*, 2025. <https://openreview.net/forum?id=bM4MbgGnpX>.
- J. Mahowald and B. C. G. Lee. "Integrating Visual and Textual Inputs for Searching Large-Scale Map Collections with CLIP." *Computational Humanities Research*, vol. 5, 2024. DOI: [10.48550/arXiv.2410.01190](https://doi.org/10.48550/arXiv.2410.01190). Best Long Paper Award.

Preprint / in review:

- J. Mahowald, B. Bell, A. Ho, M. Geyer, "Efficient Analysis of the Distilled Neural Tangent Kernel." <https://doi.org/10.48550/arXiv.2602.11320>.

Senior thesis:

- "Toward Dataset Distillation for Regression Problems." Plan II Honors Program, UT Austin, 2025. DOI: [10.26153/tsw/60640](https://doi.org/10.26153/tsw/60640).

Research projects

Scalable Verification of Physical Neural Networks

Los Alamos National Laboratory

June 2025 – Aug. 2026

Advisors: Juston Moore, Yen Ting Lin, Michael Dixon (LANL)

- Description in review.

Efficient Analysis of the Distilled Neural Tangent Kernel

Los Alamos National Laboratory

Jun. 2025 – Sep. 2026

Advisors: Brian Bell, Juston Moore (LANL)

- Developed a surrogate method to compute the neural tangent kernel using dataset distillation and random projection. Reduced computational complexity by 10^4 - 10^6 for multi-million parameter vision models while retaining >95% surrogate fidelity.
- Analyzed model redundancy in the kernel regime, characterizing calibration error, matrix conditioning, effective rank for approximated kernels, and adversarial robustness in the kernel regime.
- Introduced an algorithm for cluster-aware distillation of gradients.

Strengthening LLM guardrails using meta-learning

Los Alamos National Laboratory

Sep. – Dec. 2025

With: Michael Geyer, Selma Peterson, Nick Hengartner, Brian Bell (LANL)

- Developing a three-part framework for harmful fine-tuning and evaluation of LLM guardrail methods on several benchmarks.
- Contributing to the derivation of statistical guarantees on a language model's probability to produce harmful responses under guardrail-preserving meta-learning.

Retrieval-Augmented Search for Large-Scale Map Collections

Personal project

Aug. – Dec. 2025

With: Benjamin Lee (University of Washington)

- Extended earlier Library of Congress work by implementing the ColPali retrieval-augmented generation model, improving search precision over CLIP-based methods.
- Developed an efficient embeddings and search pipeline for 560,000+ historical maps.
- Deployed a production search engine using AWS at <http://mapras.com>.

Convergence of Dataset Distillation in Regression Problems

Applied Research Laboratories at UT Austin

Aug. 2024 – May 2025

Advisors: Atlas Wang (UT Austin), Ravi Srinivasan (ARL:UT)

- Developed dataset distillation methods for regression problems, addressing a gap in existing research which primarily focused on classification tasks.
- Reduced the operator learning problem to its regression setting to understand when and why distillation preserves relevant information. Derived sufficient near-convexity conditions under which distillation converges to an optimum, providing convergence guarantees and actionable hyperparameter selection criteria.

Generalization Limits of In-Context Operator Networks

Oden Institute for Computational Engineering & Sciences

Jun. – Aug. 2024

Advisor: Tan Bui-Thanh (UT Austin)

- Characterized empirical limits to out-of-distribution generalization of In-Context Operator Networks, a recent class of in-context learners for differential equations.
- Developed numerical data-generation schemes for high-order PDEs and systematically tested out-of-distribution performance to characterize when these networks reliably solve new problem instances.
- Trained full-parameter models on the Frontera supercomputer (TACC) using MPI and OpenMP parallel workflows.

Machine learning for archival map search

John W. Kluge Center, Library of Congress

Jan. – Apr. 2024

Advisor: Benjamin Lee (University of Washington)

- Created a zero-shot feature search engine for the Library of Congress's 560,000-item Geography & Map collection using OpenAI's CLIP transformer, enabling researchers to search maps by visual features and textual queries without manual metadata creation.
- Published a labeled dataset, fine-tuned model, and embedding library as open-source contribution.
- Received Best Long Paper award at the Computational Humanities Research conference.

Talks and presentations

- "Efficient computation of the neural tangent kernel at ImageNet-scale using dataset distillation." Laboratory Directed Research and Development (LDRD) program appraisal, Los Alamos National Laboratory. October 2025.
- "Toward efficient computation of path kernels with dataset distillation." Cybersecurity Science Research Program, Los Alamos National Laboratory. August 2025.
- "Toward dataset distillation for scientific data." Briefing to US Intelligence Community leadership. Applied Research Laboratories, University of Texas at Austin. April 2025.
- "Generalization limits of in-context operator networks for differential equations." Multi-institutional research seminar (UT Austin, Georgia Tech, Los Alamos National Laboratory). February 2025.
- "Better visual search methods with machine learning." Critical Datasets Reading Group, Library of Congress. June 2024.

Technical skills

Machine Learning: Deep learning, optimization, kernel methods, operator learning, LLM safety, compression.

Programming: Python (PyTorch, TensorFlow, JAX, Scikit-learn), parallel and distributed programming (MPI, OpenMP).

Scientific Computing: High-performance computing, numerical methods for PDEs, data analysis and visualization, MATLAB, Mathematica, Julia, C/C++.

Leadership and service

Texas Undergraduate Law Journal | Managing Editor Sep. 2021 - Apr. 2024

- Managed editorial operations and peer review process for undergraduate legal research
- Wrote analytical articles on technology policy, intellectual property, and business law

UT Student Government, Governmental Relations Agency | State Liaison Jul. 2022 - May 2023

- Organized town halls connecting UT students with Texas state legislators

References

- Brian Bell. Scientist. A-4: Advanced Research in Cyber Systems (group), Los Alamos National Laboratory. bwbell@lanl.gov.
- Juston Moore. Group leader. AI for Nuclear Deterrence (group), X Computational Physics (division), Los Alamos National Laboratory. jmoore01@lanl.gov.
- Benjamin Lee. Assistant Professor. University of Washington Information School. bcgl@uw.edu.
- William Beckner. Montgomery Professor of Mathematics. The University of Texas at Austin. beckner@math.utexas.edu.
- Ravi Srinivasan. Research scientist. Signal and Information Sciences Laboratory, Applied Research Laboratories at UT Austin. ravi.srinivasan@utexas.edu.
- Tan Bui. William J Murray Jr. Fellow in Engineering. Oden Institute for Computational Engineering and Sciences, UT Austin. tanbui@oden.utexas.edu.